

Critical Analysis: ConU: Conformal Uncertainty in Large Language Models with Correctness Coverage Guarantees

Kai-Yu Lu

1 Methodological Strengths

1.1 Problem framing and black-box realism

The study targets uncertainty quantification (UQ) in open-ended natural language generation (NLG) under black-box access, explicitly motivated by API-served large language models (LLMs) that expose only text inputs and outputs. This framing strengthens external validity because the proposed pipeline does not require logits, hidden states, or fine-tuning. The method is designed to operate entirely as post-processing over sampled generations, which aligns with the intended deployment setting.

1.2 Self-consistency-based uncertainty with explicit decomposition

The uncertainty score is grounded in a clear decomposition: (i) semantic frequency as a proxy for confidence under self-consistency, and (ii) cross-semantic consistency weighted by cluster frequency. Concretely, the generation-level uncertainty $U(\hat{y}_m^{(i)})$ (Eq. (1)) combines the frequency of the semantic cluster conveyed by a candidate and a similarity-weighted agreement term across semantic clusters. This choice is methodologically advantageous because it operationalizes an intuitive black-box signal (how often a meaning appears under sampling) while incorporating inter-cluster relations, rather than relying solely on dispersion or lexical overlap.

A second design choice strengthens interpretability for query-level uncertainty: the uncertainty of the query-response process (Eq. (2)) anchors on a representative from the largest semantic cluster, treating it as the model’s most confident semantic. This yields a consistent reference object for summarizing the uncertainty of the overall sampling process.

1.3 Correctness-aligned conformalization and explicit coverage proof

A central methodological contribution is the strict alignment of the conformal prediction (CP) nonconformity score (NS) with the uncertainty condition of correct answers. The calibration NS (Eq. (3)) selects a candidate generation that is semantically equivalent to the reference answer (via an equivalence indicator) and maximizes semantic similarity, then uses its uncertainty as r_i . This explicit coupling between correctness and the NS is a principled attempt to reduce miscalibration when logits are unavailable.

The study provides a formal coverage derivation: after computing the calibration quantile $\hat{q}$ (Eq. (4)), prediction sets are formed by filtering candidates whose uncertainty is below $\hat{q}$ (Eq. (5)), resulting in a correctness coverage guarantee $\mathbb{P}(y_{\text{test}}^* \in P(x_{\text{test}})) \geq 1 - \alpha$ (Eq. (6)) under exchangeability and a candidate-set assumption. The inclusion of this proof clarifies what is guaranteed, which assumptions are required, and how the guarantee follows from split CP.

1.4 Evaluation breadth and systematic baselines

The evaluation spans 7 instruction-tuned LLMs (including GPT-3.5-turbo and multiple LLaMA and Mistral variants) and 4 datasets covering general-purpose and medical scenarios (CoQA, TriviaQA, MedQA, MedMCQA). The experimental design compares against 8 baseline UQ methods, explicitly partitioned into white-box (Predictive Entropy, Length-normalized Predictive Entropy, Semantic Entropy, Shift Attention to Relevance) and black-box (Lexical Similarity, Number of Semantic Sets, Eccentricity, Degree Matrix). This breadth supports the claim that the proposed method is competitive across heterogeneous model families and domains.

1.5 Multi-objective reporting: discrimination, validity, and efficiency

Beyond AUROC for failure prediction, the evaluation reports (i) correctness coverage under user-specified error rates and (ii) average prediction set size, which directly correspond to CP objectives: validity and efficiency. Reporting both metrics is a methodological strength because it avoids treating UQ solely as a ranking problem and quantifies the practical cost of uncertainty sets.

1.6 Ablation studies and robustness checks

The study includes ablations on the number of sampled generations M and on the calibration:test split ratio. The analysis demonstrates that UQ performance improves with increased sampling and reports that strong AUROC can be achieved with as few as 4 generations (AUROC 0.8082 on TriviaQA with LLaMA-3-8B-Instruct). The calibration ratio analysis indicates that coverage lower bounds remain satisfied across multiple calibration sizes when using the conformal uncertainty criterion, strengthening the robustness narrative.

1.7 Selective prediction as an applied utility test

The study evaluates a downstream use-case: selecting the lowest-uncertainty generation within the calibrated prediction set to improve accuracy without fine-tuning or architectural changes. This utility test connects the conformal sets to a concrete decision rule and validates that the uncertainty signal can support post-processing beyond coverage measurement.

2 Key Limitations

2.1 Strong candidate-set assumption for correctness coverage

The correctness coverage guarantee relies on an explicit assumption that at least one sampled candidate generation satisfies the correctness criterion for each test prompt. This assumption is non-trivial in open-ended generation, especially for prompts outside model competence or under long-tail distribution. The guarantee is conditional on the success of sampling, which shifts the central challenge from “coverage given candidates” to “obtaining at least one correct candidate”.

2.2 Correctness and equivalence defined by a single similarity model and threshold

Correctness and semantic equivalence are operationalized via a cross-encoder (DistillRoBERTa backbone) sentence similarity score with a strict threshold of 0.7. This introduces several methodological risks:

- The evaluation target becomes “semantic similarity above threshold” rather than factual correctness, which can mismatch tasks that require exactness (especially in medical QA).

- The choice of the threshold 0.7 and reliance on one similarity model create sensitivity to calibration artifacts, since both NS definition (Eq. (3)) and correctness labels depend on the same similarity mechanism.
- The paper does not report sensitivity analysis for the threshold value or alternative correctness judges, limiting interpretability of the results under different evaluation criteria.

2.3 Limited task diversity and restricted NLG scope

Although the study states a focus on open-ended NLG, the evaluation datasets are QA-style tasks. The limitations section states that findings are restricted to four datasets and plans to extend to other NLG tasks such as document summarization. Consequently, conclusions about general open-ended NLG remain bounded by the tested task family.

2.4 Incomplete computational cost characterization

The approach requires sampling multiple generations per prompt (5 for UQ and 10 for coverage verification) and computing cross-encoder similarities for clustering and scoring. The paper does not report wall-clock cost, GPU usage, throughput, or scaling behavior as M grows. The absence of cost analysis limits practical assessment, particularly because similarity computation and clustering typically scale superlinearly with the number of candidates.

2.5 Empty prediction sets and deployment behavior

The study acknowledges that some test samples yield empty prediction sets after filtering by the conformal uncertainty criterion and evaluates non-empty sets separately. However, systematic analysis of the frequency of empty sets, their correlation with question difficulty, and their impact on end-to-end user experience is not provided. Deployment requires a defined behavior for empty sets (abstention, fallback generation, or additional sampling), which is not fully specified in the main pipeline.

2.6 Performance gaps and limited error analysis

The AUROC results show that the proposed method is not uniformly best across all settings. For CoQA, the average AUROC of ConU (0.7530) is slightly below Degree Matrix (0.7552). The paper does not provide an error analysis diagnosing which prompt types or semantic clustering behaviors lead to this gap. Additionally, the reported AUROC improvements on medical datasets are bounded (up to 2.4% on MedMCQA and 1.29% on MedQA), which indicates limited margin over strong baselines and motivates deeper characterization of when the method provides decisive benefits.

2.7 Scope of baseline comparison for conformalized methods

The implementation details state that two recent CP-based approaches for open-ended NLG correctness coverage are excluded from comparison due to reliance on likelihood access and inconsistent coverage across settings. While these reasons are plausible, the evaluation does not quantify the gap empirically under comparable constraints, leaving uncertainty about how much performance and efficiency are attributable to the NS alignment strategy versus general conformalization.

3 Technical Bottlenecks

3.1 Sampling and similarity computation as dominant bottlenecks

The method requires repeated generation to approximate the output distribution and requires semantic similarity computations for clustering and scoring. This introduces two bottlenecks:

- **Sampling bottleneck:** Increasing M improves UQ, but multiplies generation cost directly. This creates a cost-accuracy trade-off that constrains practical use in latency-sensitive settings.
- **Similarity bottleneck:** Cross-encoder similarity is typically expensive because it processes sentence pairs jointly. The scoring term in Eq. (1) aggregates similarities between a candidate and representatives of K clusters, and clustering typically requires pairwise comparisons among candidates. This becomes a scalability barrier as M increases or as responses lengthen.

3.2 Semantic clustering quality and stability

The uncertainty score depends on semantic clustering outcomes, which affects the frequency proxy $F(\cdot)$ and the set of cluster representatives $\hat{y}_k^{(i)}$. Clustering instability can propagate into uncertainty estimates and into CP filtering. The paper does not specify the clustering algorithm, hyperparameters, or stability diagnostics, which makes it difficult to assess how robust the approach is to clustering errors, paraphrase diversity, and near-duplicate generations.

3.3 Brittleness of correctness-aligned NS under imperfect equivalence detection

The NS definition selects a generation satisfying semantic equivalence to the reference and then maximizes similarity. If equivalence detection fails due to paraphrase mismatch, domain-specific terminology, or similarity model bias, the selected calibration NS can be mis-specified. Since CP validity depends on the empirical distribution of r_i , systematic errors in equivalence detection can shift $\hat{q}$ and induce either overly conservative sets (large sizes and frequent emptiness) or overly permissive sets (reduced practical reliability despite nominal coverage under the chosen metric).

3.4 Trade-offs among coverage, efficiency, and utility

The method aims to guarantee correctness coverage while keeping set sizes small. Empirically, at $\alpha = 0.1$, correctness coverage is at least 90% across 7 LLMs and 4 datasets, and average set sizes remain small (for example, 1.03 on TriviaQA for LLaMA-3-70B-Instruct). However, the existence of empty sets indicates that strict filtering can remove all candidates for some prompts. Selective prediction improvements depend on non-empty sets and on the quality of the uncertainty ranking. This reveals a structural trade-off: tighter thresholds increase coverage guarantees under the metric but can increase empty-set frequency and reduce usable outputs.

3.5 Exchangeability constraint and distribution shift

The formal guarantee requires exchangeability between calibration and test samples. In real-world deployment, prompts can shift over time, across user populations, or across domains, which violates exchangeability. The paper identifies extension to non-exchangeability scenarios as future work, highlighting a technical barrier to transferring the guarantee to non-stationary settings.

4 Research Implications

4.1 Black-box UQ can be strengthened by self-consistency structure

The results indicate that sampling-based self-consistency signals, when structured via semantic clustering and frequency weighting, can outperform strong baselines across multiple LLMs and datasets. This supports the implication that reliable black-box uncertainty signals exist even without logits, and that semantic-level aggregation provides a more faithful confidence proxy than lexical-level dispersion measures in open-ended generation.

4.2 Correctness coverage guarantees in open-ended NLG remain conditional on model competence

The guarantee is rigorous under stated assumptions, but the dependence on sampling at least one correct answer implies that correctness coverage is fundamentally constrained by the model’s ability to generate correct answers within a limited sample budget. This highlights a gap between statistical calibration of uncertainty and the underlying capability frontier of LLMs. Coverage guarantees become meaningful only when candidate generation has sufficient recall of correct answers.

4.3 Evaluation practices hinge on correctness definitions and judges

By using a cross-encoder similarity threshold for correctness, the study aligns with common practice in free-form QA evaluation when exact-match is too strict. The implication is that uncertainty and coverage guarantees are only as strong as the correctness judge. In domains where semantic similarity does not capture factual correctness, the reported guarantees do not translate to real-world correctness standards without redefinition of equivalence.

4.4 Conformalization provides a unifying framework for benchmarking UQ

The approach uses CP to transform heuristic uncertainty into a calibrated decision rule with explicit error control. This suggests a broader methodological direction: benchmarking UQ methods by their induced conformal sets and the resulting validity-efficiency trade-offs, rather than only AUROC. The paper’s reporting of coverage and set size alongside AUROC is aligned with this implication.

4.5 Selective prediction connects UQ to abstention and human-in-the-loop workflows

The selective prediction experiments show consistent accuracy improvements by selecting the least-uncertain generation within a calibrated set. This implies a deployment pathway: conformal sets can define a bounded candidate pool for post-processing, verification, or escalation to experts when sets are empty or uncertainty is high, aligning with abstention-based reliability systems.

5 Potential Research Directions

5.1 Guaranteeing inclusion of at least one correct candidate

A primary direction is to operationalize the paper-stated need for a criterion that verifies whether a correct answer has been sampled from the output space. Concretely:

- **Adaptive sampling policies:** Allocate generation budget based on query-level uncertainty (Eq. (2)), increasing M when the dominant semantic is weak or highly inconsistent.

- **Stopping rules under black-box constraints:** Develop likelihood-free stopping criteria based on semantic cluster stabilization (convergence of K and cluster frequencies) and uncertainty plateauing.
- **Candidate recall estimation:** Use calibration data to model the relationship between M and mis-coverage loss as formalized in the appendix, and translate it into an actionable sampling schedule for test-time prompts.

5.2 Robust correctness judges and metric-aligned guarantees

Since CP validity is defined relative to the correctness criterion, improving the correctness judge directly improves the utility of the guarantee:

- **Judge ensembles:** Combine semantic similarity with entailment-based checks or retrieval-grounded verification to reduce false positives where paraphrases hide factual errors.
- **Threshold calibration studies:** Systematically calibrate and report sensitivity of correctness coverage and set size to similarity thresholds, using calibration curves under multiple thresholds.
- **Domain-adaptive judges:** Train or select domain-specific similarity models for medical QA versus general QA to reduce judge mismatch.

5.3 Efficiency-oriented approximations for semantic clustering and similarity

To address computational bottlenecks:

- **Two-stage similarity computation:** Use embedding-based retrieval to shortlist candidate pairs, then apply cross-encoder similarity only for shortlisted pairs.
- **Cluster-level scoring:** Replace per-generation scoring with cluster-level uncertainty and then assign cluster scores to members, reducing redundant computations.
- **Approximate clustering:** Employ scalable clustering methods with explicit stability metrics and report clustering hyperparameters to improve reproducibility.

5.4 Conformal guarantees under non-exchangeability

To match deployment realities:

- **Weighted conformal prediction:** Reweight calibration samples according to similarity between calibration and test prompts to address covariate shift.
- **Online recalibration:** Maintain a rolling calibration set and update $\hat{q}$ periodically while tracking empirical coverage drift.
- **Risk-controlled conformal variants:** Extend the conformal uncertainty criterion to settings where different error types have different costs, especially in medical QA.

5.5 Expanded evaluation protocols and analyses

Several evaluation upgrades follow directly from observed gaps:

- **Empty-set characterization:** Quantify empty-set rates across datasets, models, and α , and relate them to question difficulty and cluster dispersion.
- **Failure mode analysis on CoQA:** Analyze prompts where Degree Matrix marginally outperforms ConU to isolate conditions where frequency-weighted cross-cluster similarity is insufficient.
- **Task generalization:** Extend to document summarization as stated in the limitations, and report whether semantic clustering and similarity judgments remain stable under long-form generation.
- **Cost reporting:** Report generation and similarity computation costs as functions of M and output length, enabling reproducible efficiency comparisons.

6 Conclusion

The paper presents a black-box uncertainty quantification method (ConU) that leverages self-consistency through semantic clustering and frequency-weighted cross-semantic agreement, and integrates this uncertainty into conformal prediction by defining a correctness-aligned nonconformity score. Methodologically, the work is strengthened by its realistic black-box setting, formal correctness coverage derivation, systematic comparison against eight baselines, multi-model and multi-dataset evaluation, and robustness checks across sampling budgets and calibration ratios. Empirically, ConU achieves strong AUROC on multiple datasets (for example, average 0.8224 on TriviaQA) and enables correctness coverage control across 7 LLMs and 4 datasets, while maintaining small prediction set sizes (for example, average size 1.03 on TriviaQA for LLaMA-3-70B-Instruct) and improving selective prediction accuracy.

The most critical limitations arise from the reliance on (i) the candidate-set assumption that at least one correct answer is sampled, (ii) correctness and equivalence judgments defined by a single similarity model and threshold, (iii) limited task diversity beyond QA-style datasets, and (iv) missing computational cost analysis despite sampling and similarity-intensive operations. The most promising research directions include developing black-box criteria and adaptive sampling policies that increase the probability of sampling a correct candidate, strengthening correctness judges to align guarantees with factual correctness, improving scalability of semantic clustering and similarity computation, and extending conformal validity to non-exchangeable and distribution-shifted deployment settings.